

提高任务：PID 参数调试与多关节协调

2.1 PID 参数系统化调试

对 left_shoulder_pitch 关节进行完整的 PID 参数调试，记录调试过程。

记录表格（测试至少 5 组参数）：

测试序号	Kp	Kd	效果描述（到达时间、震荡情况、稳定性）
#1			
#2			
#3			
#4			
#5			

最终参数：Kp = __, Kd = __

达到效果：_____

思考题：

- 1. 为什么在增加 Kp 值后，往往需要同步增加 Kd 值？
- 2. 如果 Kd 设置为 0，你观察到了什么现象？D 参数的作用是什么？

2.2 多关节协调控制

基于课程中的 upper_body_show.py，编写一个自定义的多关节协调动作脚本。

要求：

- 1. 同时控制至少 2 个关节实现某个动作
- 2. 使用 std_msgs/Float64MultiArray 控制多个关节，或创建多个 Publisher

提示：参考课程中的 upper_body_show.py 脚本。

截图记录：多关节协调动作的至少 2 个关键帧截图。

交付清单

2.1	1 个关节的 PID 调试记录表格 + 调试过程记录 + 思考题回答
2.2	多关节协调脚本代码 + 动作截图

提交方式：将以上所有内容整理为一份 PDF 文档，包含：

- 1. 操作步骤说明
- 2. 完整代码（复制粘贴，非截图）
- 3. 终端命令截图
- 4. Gazebo 运行效果截图
- 5. 调试记录表格

Answers:

2.1 PID 参数系统化调试

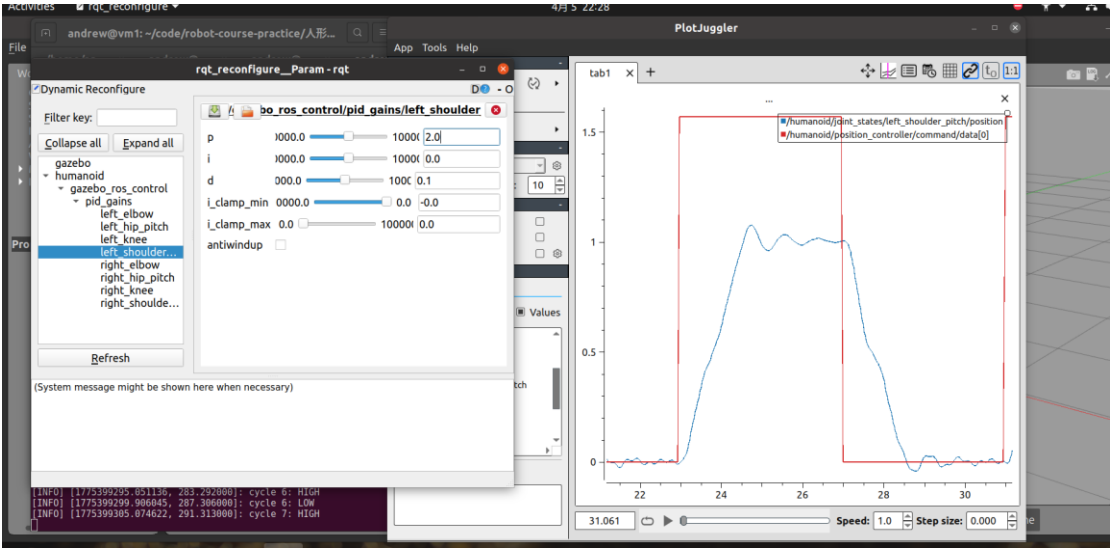
依次运行如下指令，搭建 PID 调试环境：

```
roslaunch humanoid_sim humanoid_control.launch
```

```
roslaunch rqt_reconfigure rqt_reconfigure
```

```
roslaunch plotjuggler plotjuggler
```

```
roslaunch humanoid_sim kp_square_wave_test.py
```



对 left_shoulder_pitch 关节进行 PID 参数调试，记录如下

序号	Kp	Kd	效果描述
#1	2.0	0.0	响应较慢，存在较大稳态误差，无法到达目标位置
#2	10.0	0.0	响应加快，但仍存在较大稳态误差

#3	50.0	0.0	响应更快，关节在目标位置附近小幅振荡
#4	200.0	0.0	响应更快，关节在目标位置附近小幅振荡
#5	200.0	0.5	关节在目标位置附近振幅减小
#6	200.0	2.0	振幅减小，最终收敛
#7	500.0	2.0	响应快，振荡小，最终收敛到达目标位置

最终参数：Kp = 500.0, Kd = 2.0

达到效果：关节平滑运动到目标位置，没有振荡或抖动。

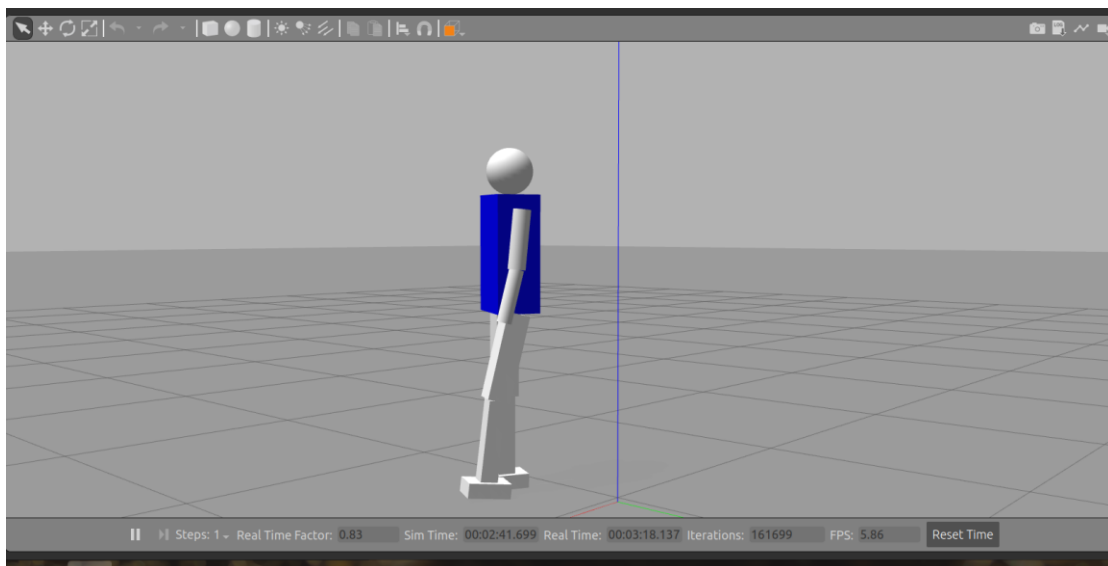
思考题：

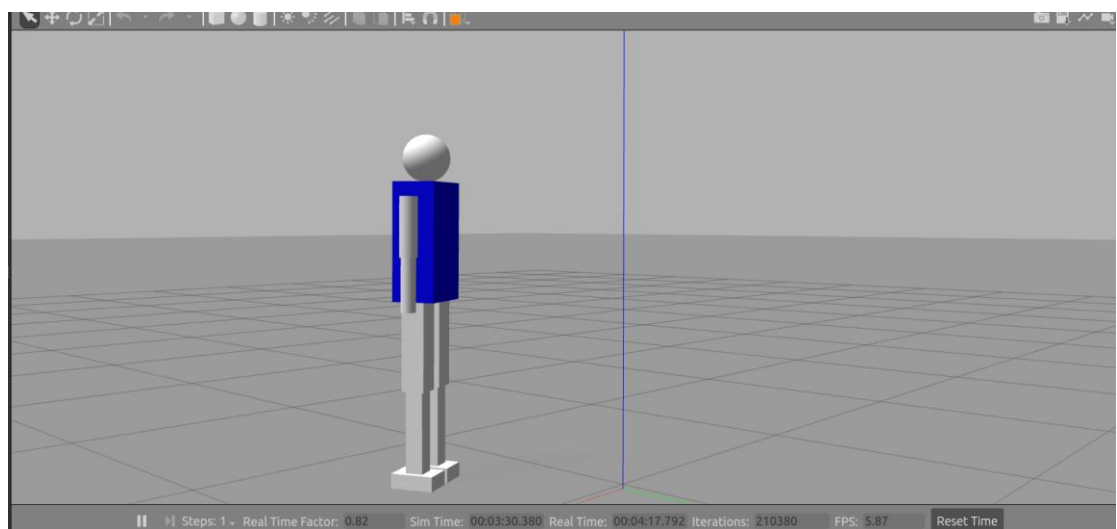
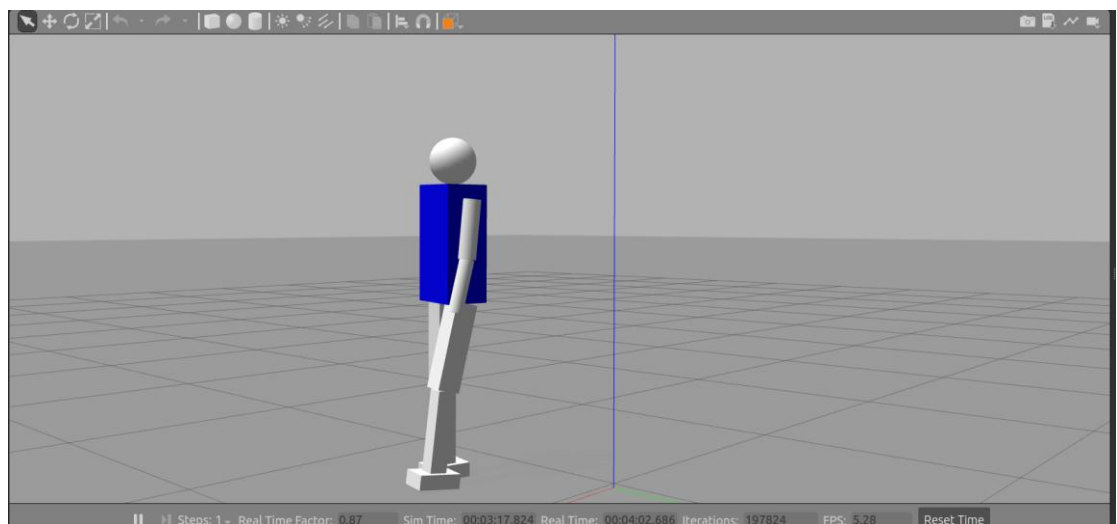
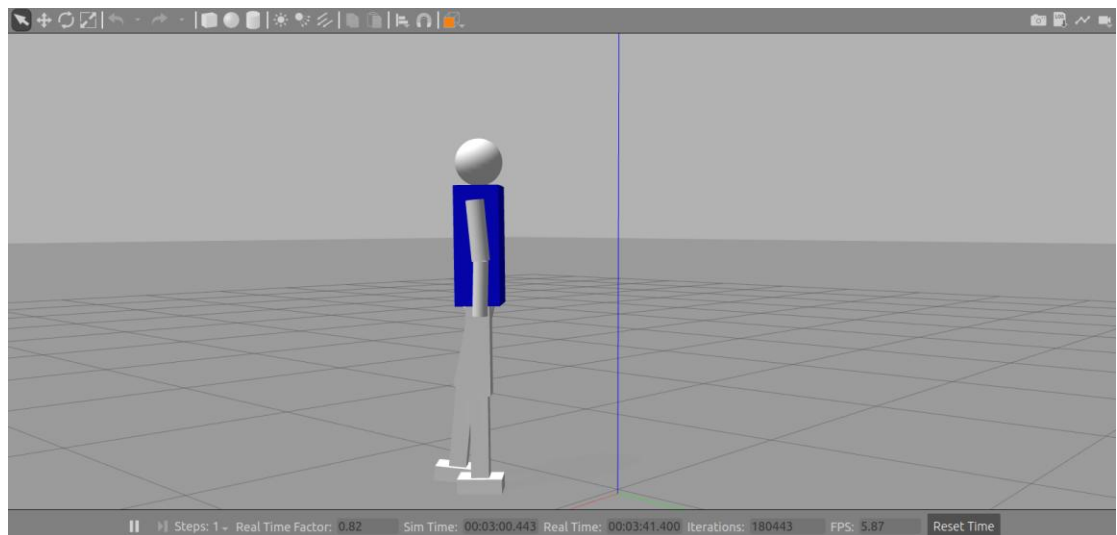
1. 为什么在增加 Kp 值后，往往需要同步增加 Kd 值？
增加 Kp 会提高系统的比例增益，加快响应速度、减小稳态误差。但过高的 Kp 会导致输出在目标值附近产生超调和振荡，同步增加 Kd 值可以利用与误差变化率成正比的阻尼作用避免超调和振荡。
2. 如果 Kd 设置为 0，你观察到了什么现象？D 参数的作用是什么？
如果 Kd 设置为 0，肩关节在抬起到目标位置后会多次上下振荡。D 参数能够抑制超调和振荡，使系统更快趋于稳定；允许使用更高的 Kp 而不引发振荡。

2.2 多关节协调控制

参照 upper_body_show.py 脚本，使用 std_msgs/Float64MultiArray 同时控制肩、肘、膝关节（髋关节随膝关节联动，防止机器人摔倒）实现机器人简单行走。

- (1) 代码参见：[walk_show.py](#)（同时附在文档最后）
- (2) 关键帧截图如下





(3) 运行日志

```

andrew@vm1:~/code/robot-course-practice$ ./walk_show.py
[INFO] [1773395022.325777, 159.305000]: WalkShow ready
[INFO] [1773395022.328380, 159.305000]: Walk show
[INFO] [1773395023.494240, 159.307000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395024.497770, 160.109000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395025.427261, 160.909000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.1, 0.1, -0.15, -0.15, 0.12, 0.0, 0.8)
[INFO] [1773395026.485178, 161.709000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395027.588679, 162.509000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395028.565542, 163.310000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
[INFO] [1773395030.834170, 165.311000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8)
[INFO] [1773395031.883720, 166.117000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.08, -0.08, -0.1, -0.1, 0.0, 0.1, 0.8)
[INFO] [1773395032.929233, 166.919000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.1, -0.1, -0.15, -0.15, 0.0, 0.12, 0.8)
[INFO] [1773395033.970167, 167.723000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.08, -0.08, -0.1, -0.1, 0.0, 0.1, 0.8)
[INFO] [1773395034.939792, 168.524000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8)
[INFO] [1773395035.919995, 169.326000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
[INFO] [1773395038.431684, 171.326000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395039.388178, 172.128000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395040.325195, 172.928000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.1, 0.1, -0.15, -0.15, 0.12, 0.0, 0.8)
[INFO] [1773395041.256815, 173.731000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395042.500737, 174.534000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395043.550477, 175.336000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
[INFO] [1773395046.008815, 177.369000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8)
[INFO] [1773395046.988385, 178.178000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.08, -0.08, -0.1, -0.1, 0.0, 0.1, 0.8)
[INFO] [1773395047.950822, 178.980000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.1, -0.1, -0.15, -0.15, 0.0, 0.12, 0.8)
[INFO] [1773395048.837594, 179.779000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.08, -0.08, -0.1, -0.1, 0.0, 0.1, 0.8)
[INFO] [1773395049.872387, 180.581000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8)
[INFO] [1773395050.921231, 181.383000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
[INFO] [1773395053.596793, 183.399000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395054.606050, 184.202000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395055.606275, 185.006000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.1, 0.1, -0.15, -0.15, 0.12, 0.0, 0.8)
[INFO] [1773395056.596368, 185.808000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395057.588104, 186.611000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395058.564571, 187.411000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
[INFO] [1773395061.087290, 189.411000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8)
[INFO] [1773395062.072802, 190.213000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.08, -0.08, -0.1, -0.1, 0.0, 0.1, 0.8)
[INFO] [1773395063.032886, 191.014000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.1, -0.1, -0.15, -0.15, 0.0, 0.12, 0.8)
[INFO] [1773395063.968937, 191.817000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.08, -0.08, -0.1, -0.1, 0.0, 0.1, 0.8)
[INFO] [1773395064.899063, 192.617000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8)
[INFO] [1773395065.814895, 193.418000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
[INFO] [1773395068.149181, 195.421000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395069.128282, 196.223000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395070.071303, 197.023000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.1, 0.1, -0.15, -0.15, 0.12, 0.0, 0.8)
[INFO] [1773395071.031025, 197.825000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.08, 0.08, -0.1, -0.1, 0.1, 0.0, 0.8)
[INFO] [1773395072.052656, 198.625000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8)
[INFO] [1773395073.002170, 199.426000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
[INFO] [1773395075.516853, 201.426000]: (left_shoulder, right_shoulder, left_elbow, right_elbow, left_knee, right_knee, duration) = (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8)

```

(4) 操作过程记录

为实现机器人行走，增加对 left_knee 和 right_knee 的设置，但调试过程中一直出现机器人摔倒的现象。联想行走时，膝关节和髋关节是联动的，简单设置髋关节的运动数值是膝关节相应的 2 倍，并不断使用 rostopic pub 一步步测试可能的 std_msgs/Float64MultiArray 值，最终确定 walk_show.py 中的 **sequence** 值。

```

/home/andrew/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人关节控制实... andrew@vm1: ~/c
std_msgs/Float64MultiArray "data: [-0.05, 0.05, -0.05, -0.05, -0.1, 0.0, 0.05, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [-0.05, 0.05, -0.05, -0.05, -0.1, 0.0, 0.05, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [-0.08, 0.08, -0.1, -0.1, -0.2, 0.0, 0.1, 0.0]" -1
publishing and latching message for 3.0 seconds
^[[Aandrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [-0.1, 0.1, -0.15, -0.15, -0.24, 0.0, 0.12, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [-0.08, 0.08, -0.1, -0.1, -0.2, 0.0, 0.1, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [-0.05, 0.05, -0.05, -0.05, -0.1, 0.0, 0.05, 0.0]" -1
publishing and latching message for 3.0 seconds
^[[A^[[Aandrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [0.05, -0.05, -0.05, -0.05, 0.0, -0.1, 0.0, 0.05]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$ rostopic pub /humanoid/position_controller/command
std_msgs/Float64MultiArray "data: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]" -1
publishing and latching message for 3.0 seconds
andrew@vm1:~/code/robot-course-practice/人形机器人综合实践课(2025年1期)/五、机器人
关节控制实践/作业5.1/workspace$

```

walk_show.py

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Walk show motion for the humanoid robot.

Features:
- One built-in Walk motion

Usage:
    roslaunch humanoid_sim humanoid_control.launch
    rosrn humanoid_sim walk_show.py
"""

import rospy
from std_msgs.msg import Float64MultiArray

JOINT_ORDER = [
    "left_shoulder_pitch",
    "right_shoulder_pitch",
    "left_elbow",
    "right_elbow",
    "left_hip_pitch",
    "right_hip_pitch",
    "left_knee",
    "right_knee",
]

class WalkShow:
    def __init__(self):
        self.index = {name: i for i, name in enumerate(JOINT_ORDER)}
        self.command = [0.0] * len(JOINT_ORDER)
        # 髋膝联动系数（用于调整质心，防止机器人摔倒）
        self.hip_scale = 2.0

        self.pub = rospy.Publisher(
            "/humanoid/position_controller/command",
            Float64MultiArray,
            queue_size=10,
        )

        rospy.sleep(1.0)
        rospy.loginfo("WalkShow ready")

    def _set_hip_knee(self):
        l_knee = self.command[self.index["left_knee"]]
        r_knee = self.command[self.index["right_knee"]]
        # 髋关节随膝关节运动（抬腿时髋关节前屈）
        self.command[self.index["left_hip_pitch"]] = (-1) *
self.hip_scale * l_knee
        self.command[self.index["right_hip_pitch"]] = (-1) *
self.hip_scale * r_knee

    def _publish(self):
        self._set_hip_knee()
        self.pub.publish(Float64MultiArray(data=self.command))

# 再添加左右膝参数
```

```

def walk(self, l_shoulder, r_shoulder, l_elbow, r_elbow, l_knee,
r_knee, duration=0.8, rate_hz=30.0):
    duration = max(0.05, duration)
    rate_hz = max(10.0, rate_hz)

    target = list(self.command)
    target[self.index["left_shoulder_pitch"]] = l_shoulder
    target[self.index["right_shoulder_pitch"]] = r_shoulder
    target[self.index["left_elbow"]] = l_elbow
    target[self.index["right_elbow"]] = r_elbow
    target[self.index["left_knee"]] = l_knee
    target[self.index["right_knee"]] = r_knee

    steps = max(1, int(duration * rate_hz))
    start = list(self.command)
    rate = rospy.Rate(rate_hz)

    for step in range(1, steps + 1):
        alpha = float(step) / float(steps)
        for i in range(len(self.command)):
            self.command[i] = start[i] + alpha * (target[i] -
start[i])
        self._publish()
        rate.sleep()

def home(self, duration=1.0):
    self.walk(0.0, 0.0, 0.0, 0.0, 0.0, 0.0, duration=duration)

def run_show(self):
    rospy.loginfo("Walk show")
    self.home(duration=1.0)

sequence = [
    # (left_shoulder, right_shoulder, left_elbow, right_elbow,
left_knee, right_knee, duration)
    # 左前
    (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8),
    (-0.08, 0.08, -0.1, -0.1, 0.10, 0.0, 0.8),
    (-0.1, 0.1, -0.15, -0.15, 0.12, 0.0, 0.8),
    (-0.08, 0.08, -0.1, -0.1, 0.10, 0.0, 0.8),
    (-0.05, 0.05, -0.05, -0.05, 0.05, 0.0, 0.8),
    # 恢复站立
    (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2),
    # 右前
    (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8),
    (0.08, -0.08, -0.1, -0.1, 0.0, 0.10, 0.8),
    (0.1, -0.1, -0.15, -0.15, 0.0, 0.12, 0.8),
    (0.08, -0.08, -0.1, -0.1, 0.0, 0.10, 0.8),
    (0.05, -0.05, -0.05, -0.05, 0.0, 0.05, 0.8),
    # 恢复站立
    (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2)
]

for _ in range(4):
    for pose in sequence:
        rospy.loginfo("(left_shoulder, right_shoulder,
left_elbow, right_elbow, left_knee, right_knee, duration) = %s", pose)
        self.walk(*pose)

    self.home(duration=1.0)

def main():

```

```
    rospy.init_node("walk_show", anonymous=True)
    show = WalkShow()
    show.run_show()

if __name__ == "__main__":
    try:
        main()
    except rospy.ROSInterruptException:
        pass
```